Unit - II Force and Force System

Introduction to Mechanics

MODULE - 1

Mechanics is a physical science Concerned with the study of response of bodies under the application of forces.

Engineering Mechanics is the application of Mechanics to the solution of Engineering Problems.

It is broadly classified into three types

- Mechanics of Rigid bodies
- Mechanics of Deformable bodies
- Mechanics of Fluids

(1) Mechanics of Rigid bodies :

It is the branch of science which deals with the study of bodies that donot undergo any deformation under the application of forces.

It can further be classified into Statics & Dynamics.

Statics

It is the branch of mechanics which deals with the study of the behaviour of bodies of Particles in the state of rest.

Dynamics

It is the branch of mechanics, which deals with the study of the behaviour of bodies of Particles in the state of motion.

Dynamics can be further classified into

Kinematics :- The forces causing the motion are not considered.

Kinetics :- The forces causing the motion are mainly considered.

(2) Mechanics of Deformable bodies :

It is the branch of science which deals with the stu
of bodies that undergo deformation under the applica
of forces.

It can be classified into ⓐ Strength of materials
ⓑ Theory of plasticity
ⓒ Theory of Elasticity

(3) Mechanics of fluids :

It is the branch of science which deals with the stu
of fluids.

Basic Idealisations :

① Particle :

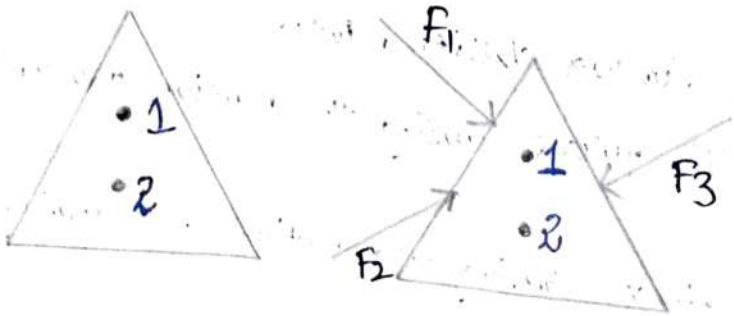
A body of infinitely small volume whose mass can be neglected is called a particle.

② Body :

The assemblage of a number of particles is known as a bo

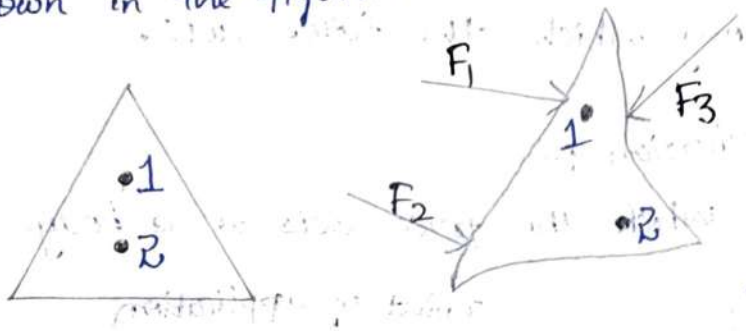
③ Rigid Body :

A Rigid Body is one in which the positions of the constituent particles do not change under the application of external for
such as the positions of Particles 1 & 2 as in the figure.



④ Deformable Body:

The Deformable Body is one in which the positions of the constituent particles change under the application of external forces, such as the positions of particles 1 & 2 as shown in the figure.



⑤ Continuum:

The fragmentation of matter into molecules, atoms, electrons etc is too complex and as such it is not required for many Problems in Engineering Mechanics. In most problems, an Engineer is interested only in overall measurable manifestation like Pressure, Density and Temperature that have gross effects on the actions on many molecules & atoms which can be imagined to arise from a continuous distribution of matter, which is known as Continuum.

Force

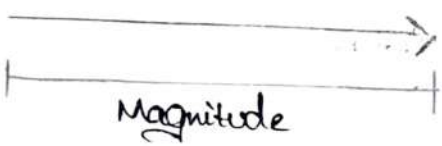
A force is an external agency which tends to change the state of a body of a particle. When a force is applied to a body which is at rest, the body may remain in the state of rest of it may move with some velocity.

The S.I units of force is Newtons (N).

Elements of a force / Characteristics of a force.

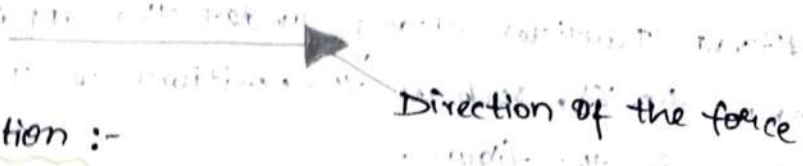
(1) Magnitude :—

The length of the vector represents the magnitude of the force.



(2) Direction :-

The direction of the force can be represented by an arrow head.

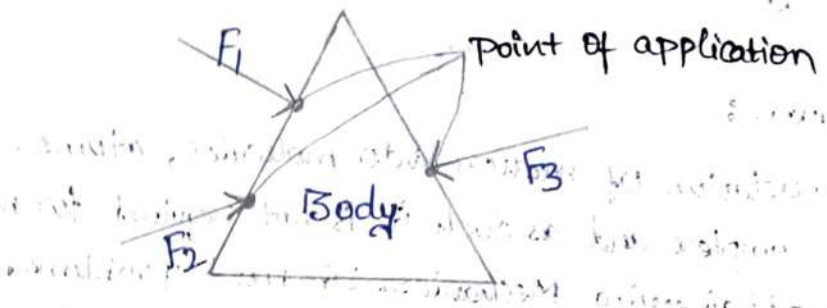


(3) Line of Action :-

It is the line along which the force acts.

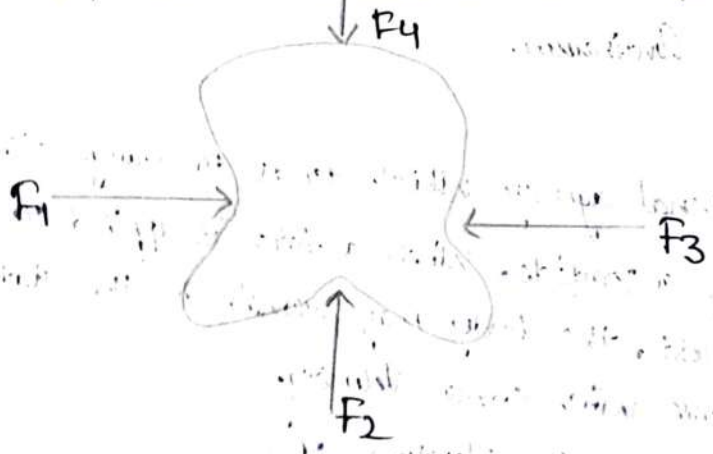
(4) Point of application :-

It is a point at which the force acts on a body.



Force System

If two or more forces are acting on a body or a particle, then it is said to be force system.



Types of force system

1. Coplanar Force system
2. Non-coplanar Force system
3. Collinear Force system
4. Non - collinear Force system

(1) Coplanar Force system

If two or more forces are acting in a single plane, then it is said to be a coplanar force system.

The types of coplanar force system are:

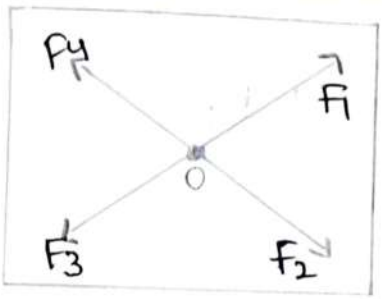
(i) Coplanar Concurrent force system

(ii) Coplanar non-Concurrent force system

(iii) Coplanar Parallel force system

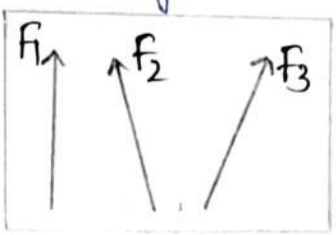
(i) Coplanar Concurrent Force system :-

If two or more forces are acting in a single plane and their line of action pass through single point then it is said to be coplanar concurrent force system.



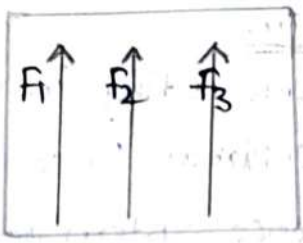
(ii) Coplanar non-concurrent Force system :-

If two or more forces are acting in a single plane and their line of action do not meet at a common point then the forces constitute a coplanar non-concurrent forcesystem.



(iii) Coplanar Parallel force System :-

If two or more forces acting in a single plane with their lines of action Parallel to one another constitute a coplanar Parallel force system.

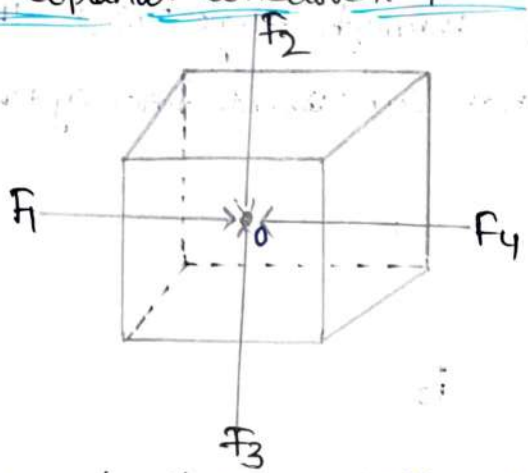


② Non-Coplanar Force System:

If two or more forces acting in different planes, the forces constitute a Non-Co-planar force system.

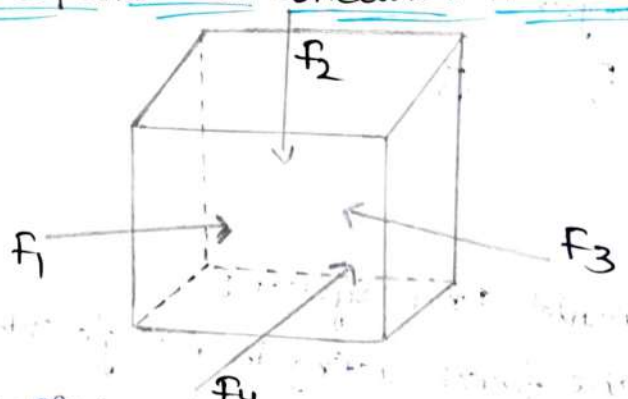
Such a system of forces can be:

(i) Non-Coplanar Concurrent force system :-



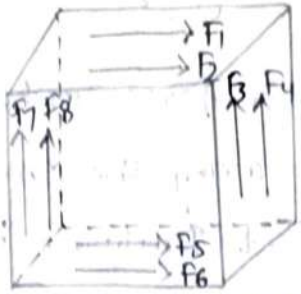
If a system of forces acting on different planes but Pass through the same point, then it is said to be Non-coplanar concurrent force system.

(ii) Non-Coplanar non-Concurrent force system :-



If two or more forces are acting on different planes do not pass through the same point, they constitute a Non-Coplanar non-Concurrent force system.

(iii) Non-Coplanar Parallel Force System :-



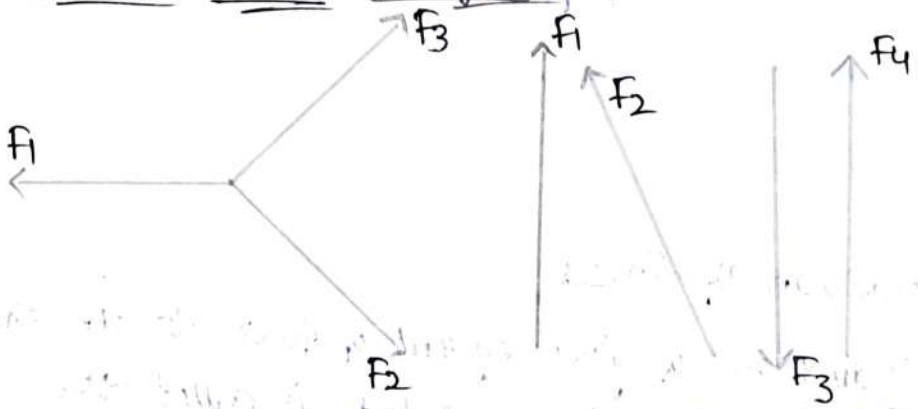
If two or more forces acting in different planes and are parallel to one another, the system of force is said to be Non-Coplanar Parallel Force System.

(3) Collinear Force System

If the lines of actions of two or more forces coincide with one another it constitutes a Collinear force system.



(4) Non-Collinear Force System

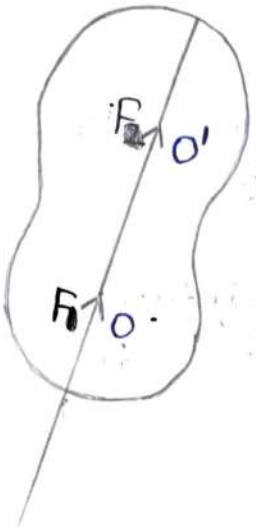


If the lines of actions of forces do not coincide with one another it is called as Non-collinear force system.

Principle of Transmissibility of Forces

This principle states that a force can be transmitted from one point to another along the same line of action such that the effect produced by the force on a body remains unchanged.

Let us consider a rigid body subjected to a force of 'F' at point 'O' as shown in figure according to the Principle of Transmissibility, the force 'F' can be transmitted to a new point 'O1' along the same line of action such that the net effect remains unchanged.



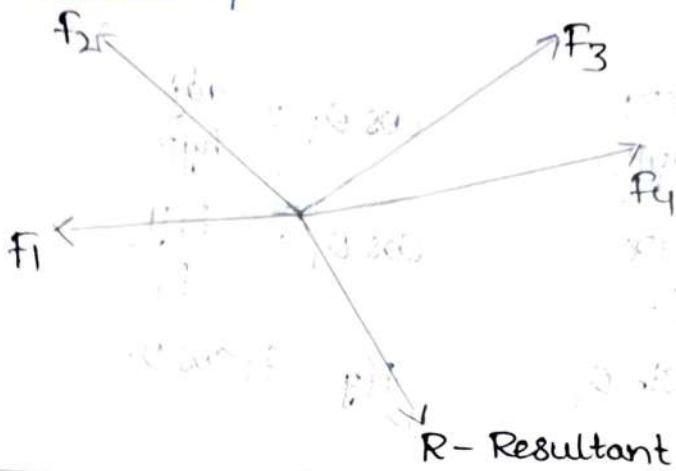
Composition of Forces

The reduction of given system of forces to the simplest system that will be its equivalent is called the Composition of forces.

If several forces like F1, F2, F3, etc are applied to a body at one point and all the forces acting in the same plane which represent a system of forces that can be reduced to a single resultant force. This process is called Composition of forces.

Resultant

If number of forces F_1, F_2, F_3 etc are acting on a particle, it is possible to find out a single force which would replace them i.e, which produces the same effect as produced by all the forces. This single force is called Resultant of Resultant force ('R').

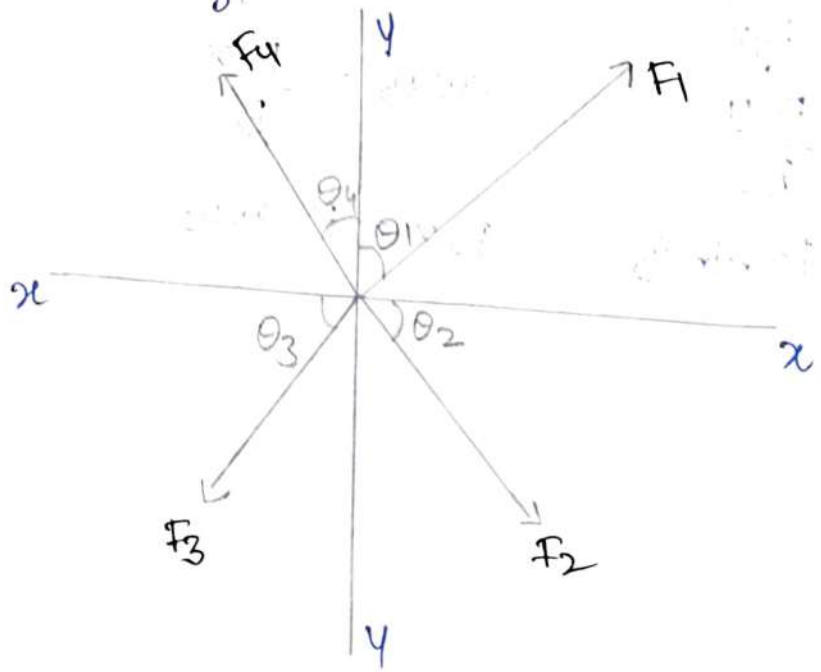


$$R = \sqrt{F_x^2 + F_y^2}$$

$$\theta = \tan^{-1}\left[\frac{F_y}{F_x}\right]$$

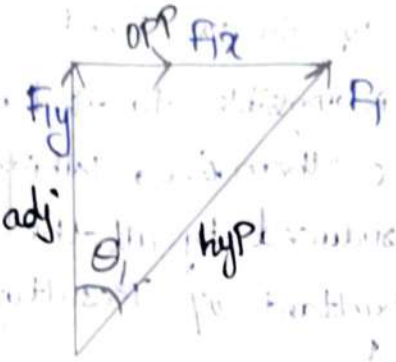
Resolution of forces

It is the process of splitting up of force into number of components without changing its effect on the body. The resolution of the force is necessary if the force is inclined at an certain angle with respect to either vertical or horizontal direction.



Force 1 [F1]:



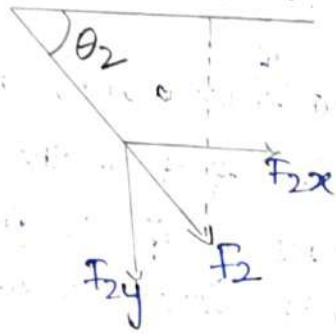


$$\sin\theta_1 = \frac{opp}{hyp} \qquad \cos\theta_1 = \frac{adj}{hyp}$$

$$\sin\theta_1 = \frac{F_{1x}}{F_1} \qquad \cos\theta_1 = \frac{F_{1y}}{F_1}$$

$$F_{1x} = F_1 \sin\theta_1 \qquad F_{1y} = F_1 \cos\theta_1$$

Force 2 [F2]:



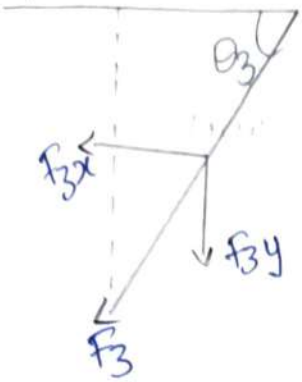


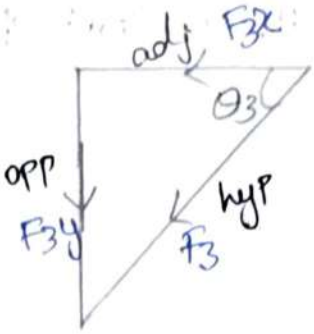
$$\sin\theta_2 = \frac{opp}{hyp} \qquad \cos\theta_2 = \frac{adj}{hyp}$$

$$\sin\theta_2 = \frac{F_{2y}}{F_2} \qquad \cos\theta_2 = \frac{F_{2x}}{F_2}$$

$$F_{2y} = F_2 \sin\theta_2 \qquad F_{2x} = F_2 \cos\theta_2$$

Force 3 [F3]:





$$\sin\theta_3 = \frac{opp}{hyp}$$

$$\sin\theta_3 = \frac{F_{3y}}{F_3}$$

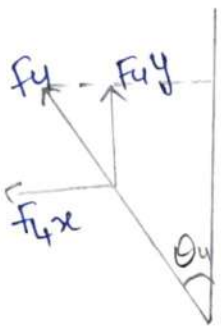
$$F_{3y} = F_3 \sin\theta_3$$

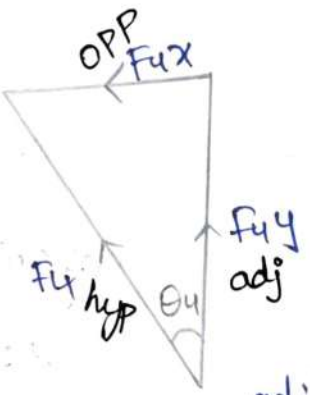
$$\cos\theta_3 = \frac{adj}{hyp}$$

$$\cos\theta_3 = \frac{F_{3x}}{F_3}$$

$$F_{3x} = F_3 \cos\theta_3$$

Force 4 [F4]:





$$\sin\theta_4 = \frac{opp}{hyp}$$

$$\sin\theta_4 = \frac{F_{4x}}{F_4}$$

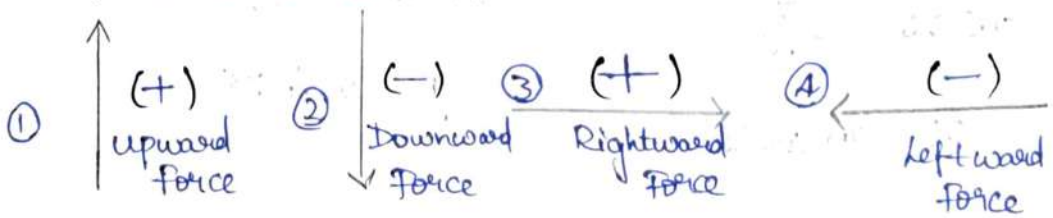
$$F_{4x} = F_4 \sin\theta_4$$

$$\cos\theta_4 = \frac{adj}{hyp}$$

$$\cos\theta_4 = \frac{F_{4y}}{F_4}$$

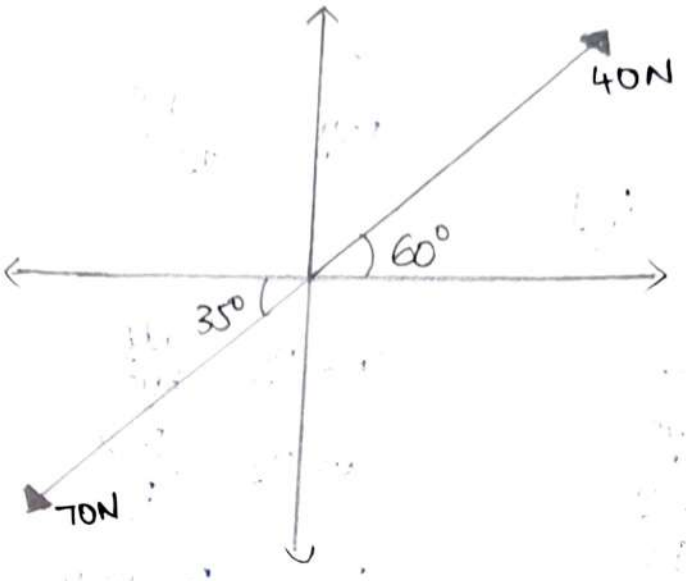
$$F_{4y} = F_4 \cos\theta_4$$

Sign Convention

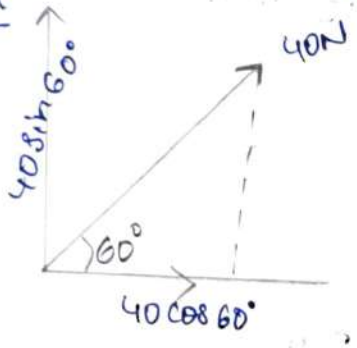


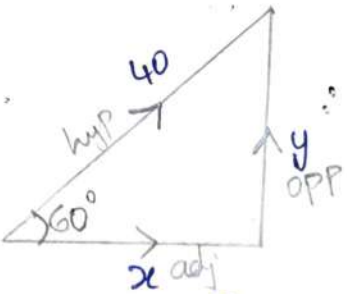
Problems

1) Determine the resultant of the following force system.



Sol



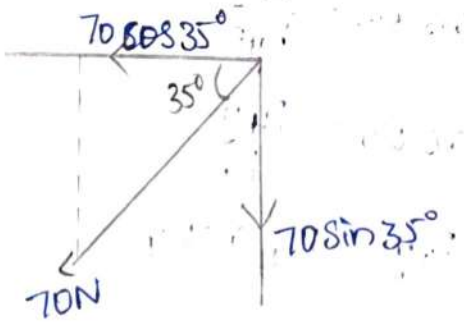


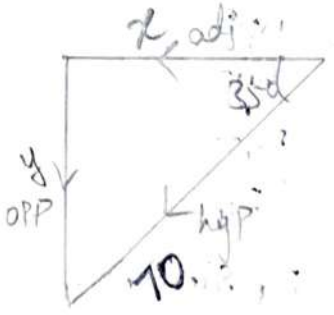
$\sin 60° = \frac{y}{40}$

$y = 40 \sin 60°$

$\cos 60° = \frac{x}{40}$

$x = 40 \cos 60°$





$\cos 35° = \frac{x}{70}$

$x = 70 \cos 35°$

$\sin 35° = \frac{y}{70}$

$y = 70 \sin 35°$

$$R = \sqrt{\Sigma H^2 + \Sigma V^2}$$

$$\Sigma H = 40\cos 60^\circ - 70\cos 35^\circ$$

$$\boxed{\Sigma H = -37.34\ N}$$

$$\Sigma V = 40\sin 60^\circ - 70\sin 35^\circ$$

$$\boxed{\Sigma V = -5.50\ N}$$

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$R = \sqrt{(-37.34)^2 + (-5.50)^2}$$

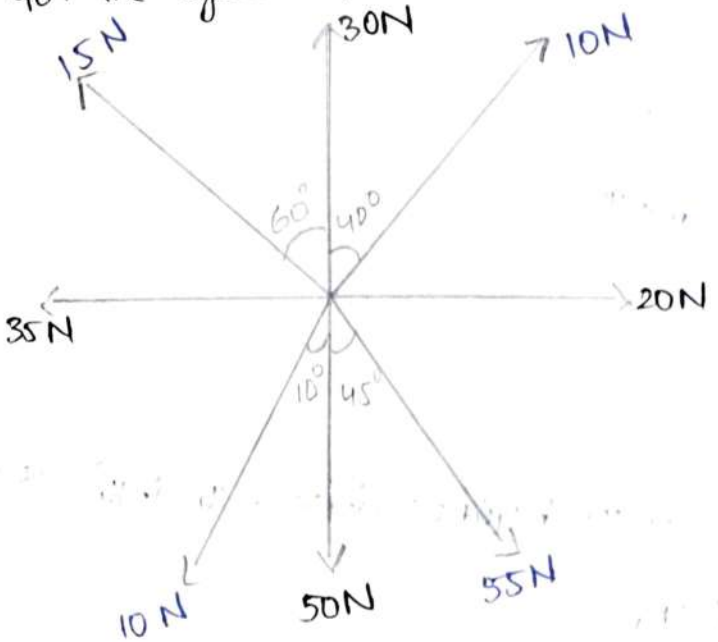
$$\boxed{R = 37.75\ N}$$

$$\theta = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$$

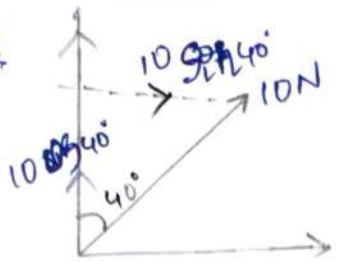
$$\theta = \tan^{-1}\left[\frac{-5.50}{-37.34}\right]$$

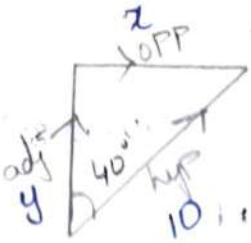
$$\boxed{\theta = 8.379^\circ}$$

2) Calculate magnitude and direction of the resultant force for the system of forces shown in the figure.



Sol



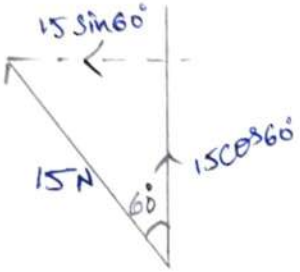


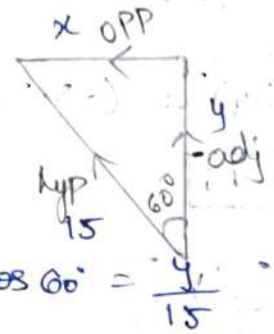
$\sin 40^\circ = \frac{x}{10}$

$x = 10 \sin 40^\circ$

$\cos 40^\circ = \frac{y}{10}$

$y = 10 \cos 40^\circ$



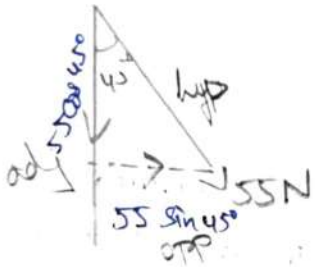


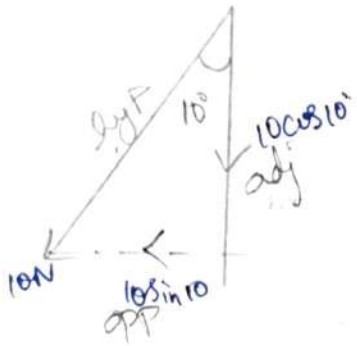
$\sin 60^\circ = \frac{x}{15}$

$x = 15 \sin 60^\circ$

$\cos 60^\circ = \frac{y}{15}$

$y = 15 \cos 60^\circ$





$\Sigma H = -35 + 20 + 10 \sin 40^\circ + 55 \sin 45^\circ - 10 \sin 10^\circ - 15 \sin 60^\circ$

$\boxed{\Sigma H = 15.59 N}$

$\Sigma V = 30 - 50 + 10\cos 40^\circ - 55\cos 45^\circ - 10\cos 10^\circ + 15\cos 60^\circ$

$\boxed{\Sigma V = -53.57\,N}$

$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$

$R = \sqrt{(15.59)^2 + (-53.57)^2}$

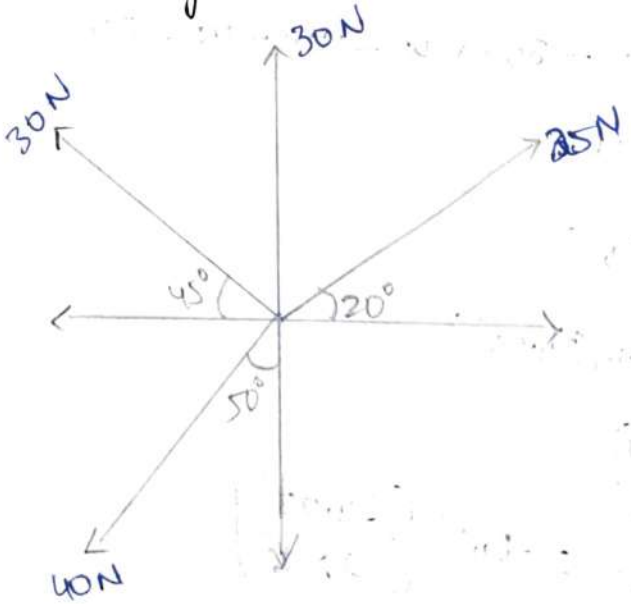
$\boxed{R = 55.79\,N}$

$\theta = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$

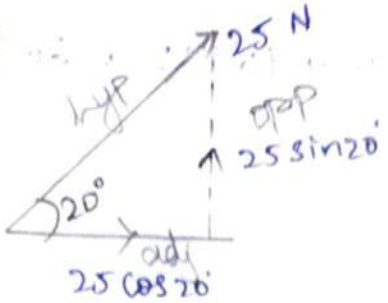
$\theta = \tan^{-1}\left[\frac{-53.57}{15.59}\right]$

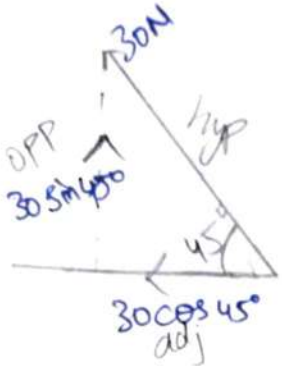
$\boxed{\theta = -73.77^\circ}$

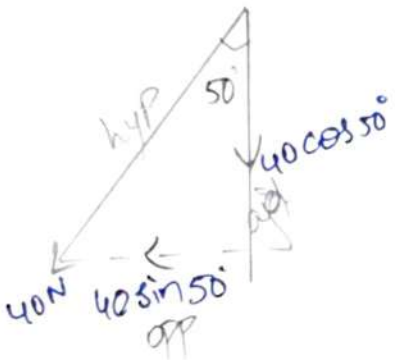
3) Find the magnitude and direction of the resultant force for the figure shown below.



Sol







$$\Sigma H = 25\cos 20^\circ - 30\cos 45^\circ - 40\sin 50^\circ$$

$$\boxed{\Sigma H = -28.36\ N}$$

$$\Sigma V = 25\sin 20^\circ + 30\sin 45^\circ - 40\cos 45^\circ + 30$$

$$\boxed{\Sigma V = 34.05\ N}$$

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

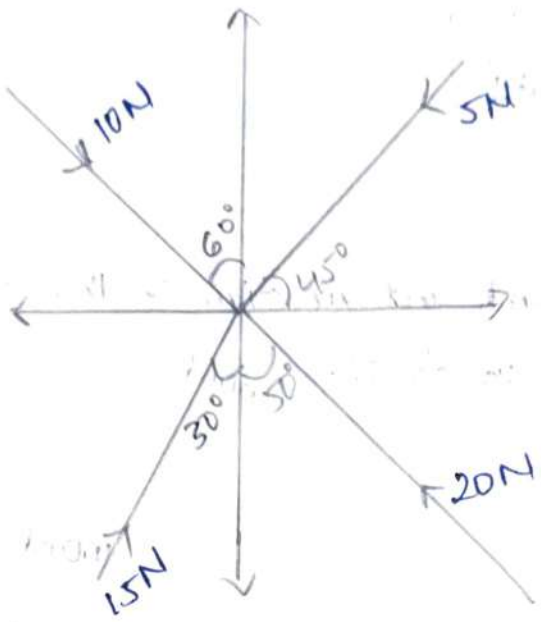
$$R = \sqrt{(-28.36)^2 + (34.05)^2}$$

$$\boxed{R = 44.31\ N}$$

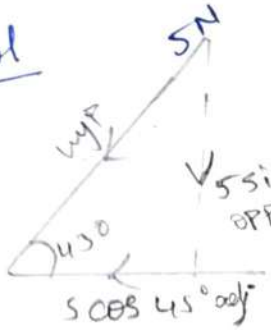
$$\theta = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right] = \tan^{-1}\left[\frac{34.05}{-28.36}\right]$$

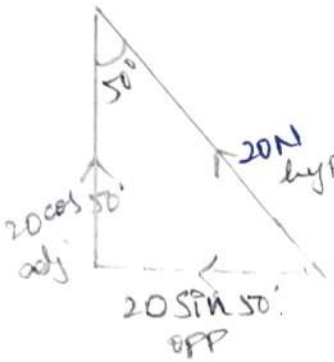
$$\boxed{\theta = -50.20^\circ}$$

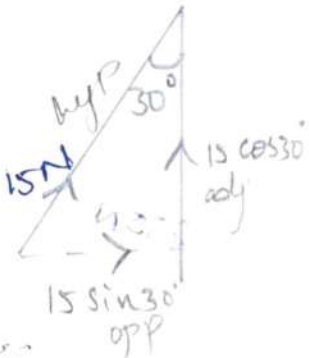
4) find the magnitude and direction, for the given system of forces. of the resultant



Sol









$$\Sigma H = -5\cos 45^\circ - 20\sin 50^\circ + 15\sin 30^\circ + 10\sin 60^\circ$$

$$\boxed{\Sigma H = -2.69\ N}$$

$$\Sigma V = -5\sin 45^\circ + 20\cos 50^\circ + 15\cos 30^\circ - 10\cos 60^\circ$$

$$\boxed{\Sigma V = 17.31\ N}$$

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$R = \sqrt{(-2.69)^2 + (17.31)^2}$$

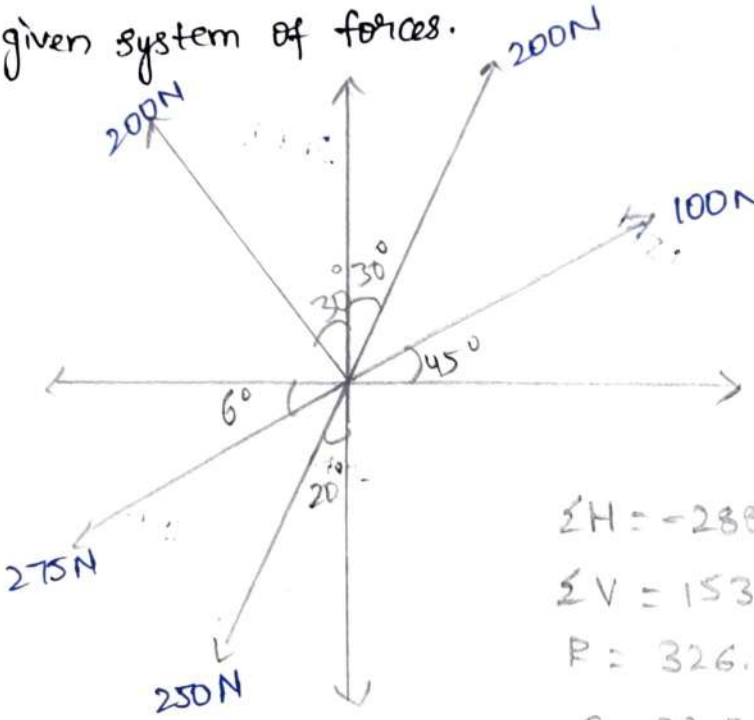
$$\boxed{R = 17.50\ N}$$

$$\theta = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$$

$$\theta = \tan^{-1}\left[\frac{17.31}{-2.69}\right]$$

$$\boxed{\theta = -81.16^\circ}$$

5) Find the resultant and direction of the resultant for the given system of forces.



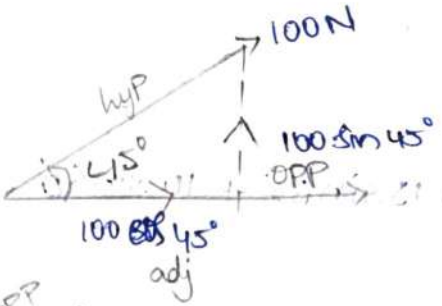
$\Sigma H = -288.287\ N$

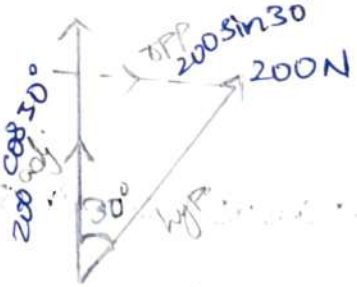
$\Sigma V = 153.453\ N$

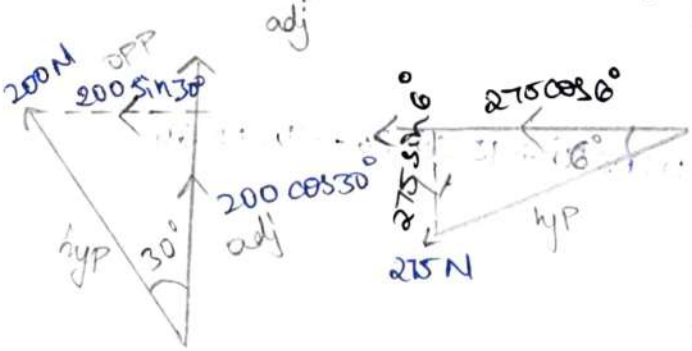
$P = 326.584\ N$

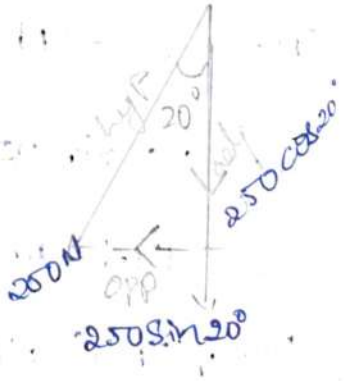
$\theta = 28.02^\circ$

Sol









$\Sigma H = 100\cos 45^\circ + 200\sin 30^\circ - 200\sin 30^\circ - 275\cos 6^\circ - 250\sin 20^\circ$

$\boxed{\Sigma H = -288.287 N}$

$\Sigma V = 100\sin 45^\circ + 200\cos 30^\circ + 200\cos 30^\circ - 275\sin 6^\circ - 250\cos 20^\circ$

$\boxed{\Sigma V = 153.452 N}$

$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$

$R = \sqrt{(-288.287)^2 + (153.452)^2}$

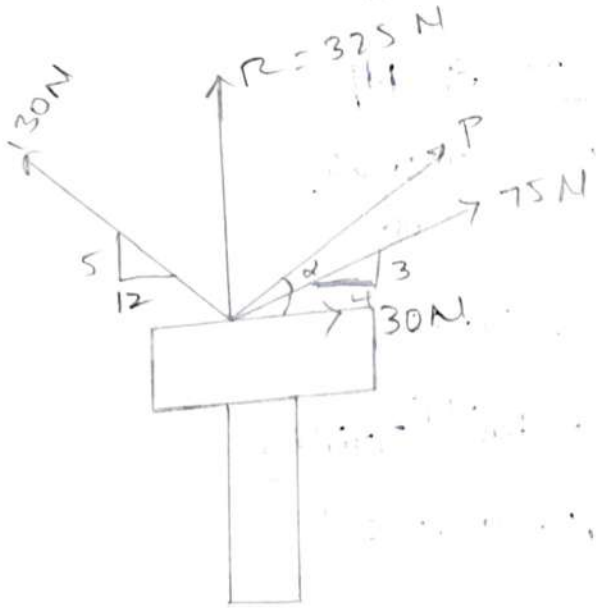
$\boxed{R = 326.583 N}$

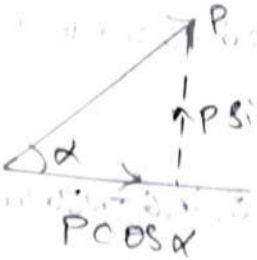
$\theta = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$

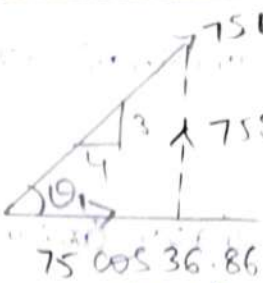
$\theta = \tan^{-1}\left[\frac{153.452}{-288.287}\right]$

$\boxed{\theta = -28.02^\circ}$

6) A boat is subjected to the forces as shown in the figure. Determine the magnitude of direction of the unknown, Sources required to the apply at Point A, So that the boat is drawn out with a net resultant vertical upward vertical force;

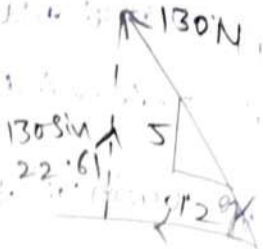






$\theta_1 = \tan^{-1}\left[\frac{3}{4}\right]$

$\theta_1 = 36.86°$



$\theta_2 = \tan^{-1}\left[\frac{5}{12}\right]$

$\theta_2 = 22.61°$

$\Sigma H = 0$

$75 \cos 36.86 + P\cos\alpha - 130 \cos 22.61 + 30 = 0$

$$P\cos\alpha = 30 \longrightarrow (1)$$

$\Sigma V = 325$

$75 \sin 36.86 + P\sin\alpha + 130 \sin 22.61 = 325$

$$P\sin\alpha = 230.03 \longrightarrow (2)$$

Squaring and adding (1) and (2)

$P^2\cos^2\alpha + P^2\sin^2\alpha = (30)^2 + (230.03)^2$

$P^2[\cos^2\alpha + \sin^2\alpha] = 53813.80$

$P^2 = \sqrt{53813.80}$

$$P = 231.97\,N$$

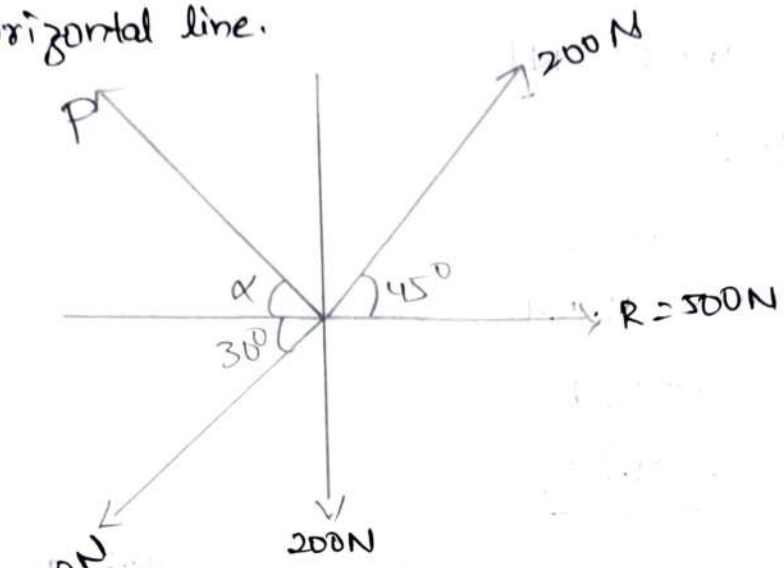
$\frac{(2)}{(1)} \Rightarrow \frac{P\sin\alpha}{P\cos\alpha} = \frac{230.03}{30}$

$\tan\alpha = 7.668$

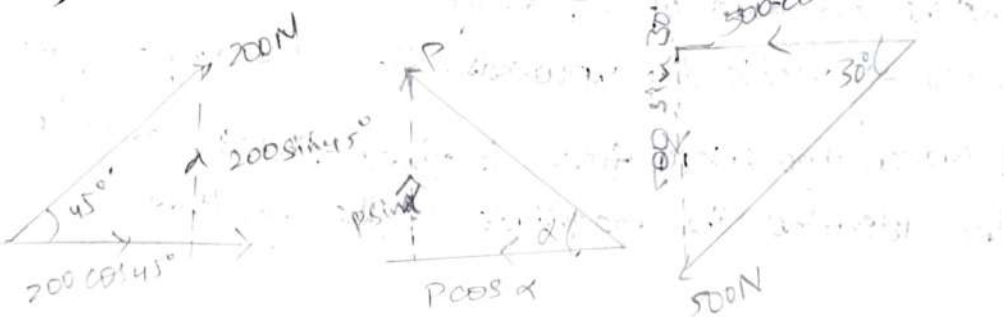
$\alpha = \tan^{-1}[7.668]$

$$\alpha = 82.56$$

7) four coplanar forces acting at a point are shown in the figure below. One of the forces is unknown and its magnitude is P. The resultant has a magnitude of 500N and is acting along x-axis. Find the unknown force P and its inclination with respect to horizontal line.



Sol



$\Sigma H = 500$

$200\cos 45 - P\cos\alpha - 500\cos 30^\circ = 500$

$P\cos\alpha = 200\cos 45 - 500\cos 30 - 500$

$P\cos\alpha = -791.59\,N \rightarrow (1)$

$\Sigma V = 0$

$200\sin 45^\circ + P\sin\alpha - 500\sin 30 - 200 = 0$

$P\sin\alpha = 200 + 500\sin 30 - 200\sin 45^\circ$

$P\sin\alpha = 308.57 \rightarrow (2)$

Squaring → adding. ① → ②

$$P^2[\sin^2\alpha + \cos^2\alpha] = (-791.59)^2 + (308.57)^2$$

$$P^2 = 721830.173$$

$$P = \sqrt{721830.173}$$

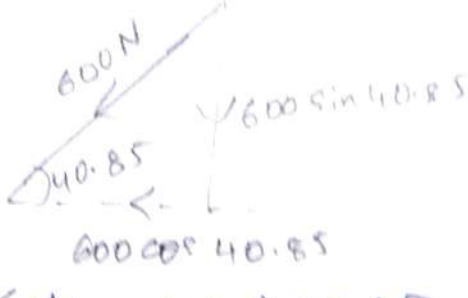
$$\boxed{P = 849.60 \text{ N}}$$

Eqn ② ÷ ①

$$\frac{P\sin\alpha}{P\cos\alpha} = \frac{308.57}{-791.59}$$

$$\tan\alpha = -0.389 \Rightarrow \alpha = \tan^{-1}(-0.389) \Rightarrow \alpha = -21.25$$

$$\boxed{\alpha = -21.29}$$



$\Sigma V = -600 \sin 40.85$

$\Sigma V = -392.44 N$

$\Sigma H = 800 - 600 \cos 40.85$

$\Sigma H = 346.14 N$

$R = \sqrt{(\Sigma V)^2 + (\Sigma H)^2}$

$R = \sqrt{(-392.44)^2 + (346.14)^2}$

$\boxed{R = 523.28 N}$

$\alpha = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$

$\alpha = \tan^{-1}\left[\frac{-392.44}{346.14}\right]$

$\boxed{\alpha = -48.58^\circ}$

MODULE - 2

Moment of a force

The rotational effect produced by a force is known as Moment.

Its unit is Nm or Nmm or KN-m or KN-mm.

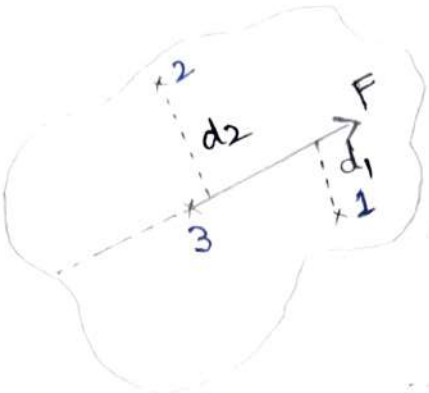
Example :-

(a) Opening or closing the door along a fixed hinge.

(b) A see-saw

(c) Unscrewing a nut with a spanner.

Moment is defined as the product of force and a perpendicular distance between the line of the action of the fore and the moment of center.



The Point about which moment is considered is Called Moment center.

As per the above figure 1, 2, 3 are called as Moment centers.

The perpendicular distance of the Point from the line of action of the force is called Moment arm.

As per the above figure d_1, d_2 are called as Moment arms.

The moment of force F about Point 1 from the line the action of the force is given by $M_1 = F \times d_1$

Similarly, moment about Point 2 is given by $M_2 = F \times$

If the moment center 3 lies on the line of action of the force 'F', the moment arm is zero (0) and hence it given as $M_3 = F \times 0 = 0$

The moment of the force has got direction also.

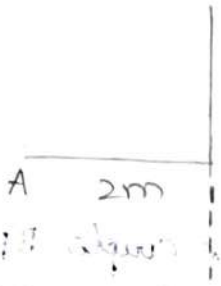
In the above figure M_1 is clock-wise [+ moment] M_2 is Anti clock-wise [- moment].

Types of Moment

There are two types of moment namely,

(1) Clockwise moment

If a body rotates clockwise with respect to the force, the moment is known as clockwise Moment and it is positive (+ve) moment.



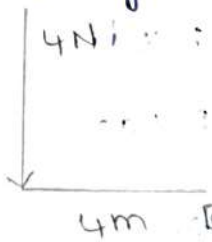
Moment about point 'A' $= M_A$

$= 2 \times 2$

$= 4\ KN-m$

(2) Anticlockwise Moment

If a body rotates anticlock wise with respect to the force, the moment is known as, Anticlock wise Moment and it is Negative (-ve) moment.



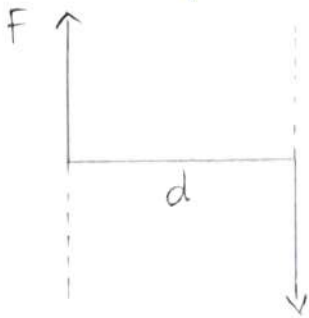
Moment about point 'B' $= M_B$

$= 4 \times 4$

$= -16\ N-m$

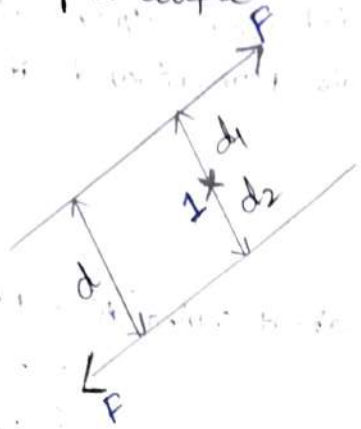
Couple:

Two Parallel forces equal in magnitude and opposite in direction and are seperated by a definite distance are said to form a Couple.



Eg: Steering Wheel, Opening and closing a tap, Opening and closing the cap of a bottle, Winding a watch

Moment of a Couple



Let the magnitude of the forces forming the couple be F and the perpendicular distance between the two forces be d.

Consider the moments of two forces constituting a couple about '1' as shown in the above figure.

Moment of upper force about '1' = $F \times d_1$

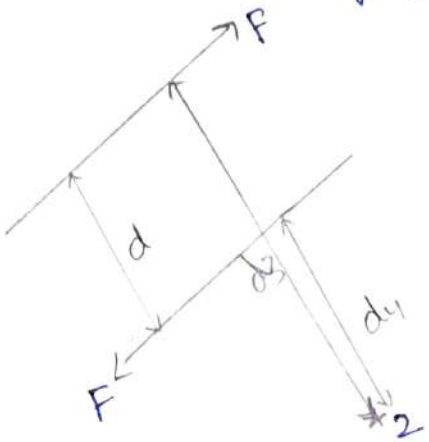
Moment of lower force about '1' = $F \times d_2$

Moment of Couple = $F \times d_1 + F \times d_2$

$= F[d_1 + d_2]$

$= Fd$

Now consider the moment of the forces about point '2' which is outside the two forces as shown in the figure



Moment of upper force about '2' = $F \times d_3$

Moment of lower force about '2' = $-F \times d_4$

Moment of couple = $F \times d_3 - F \times d_4$

$= F[d_3 - d_4]$

$= Fd$

Characteristics of Couple :

→ A couple consists of a pair of equal and opposite parallel forces which are seperated by a definite distance.

→ The translatory effect of a couple on the body is zero (0).

→ The rotational effect [moment] of a couple about any point is a constant and it is equal to the product of the magnitude of the forces and the perpendicular distance between the two forces.

→ The effect of the couple is unchanged if

(a) The couple is rotated through any angle.

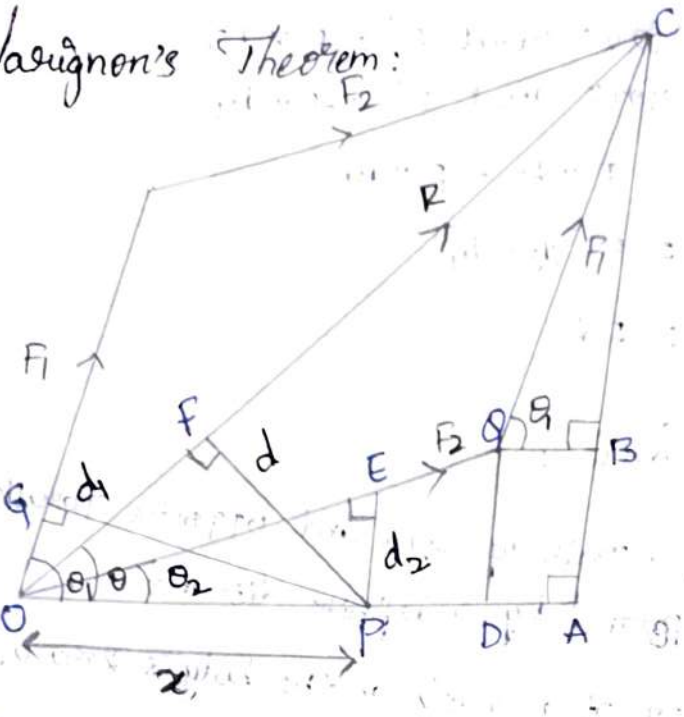
(b) The couple is shifted to any other position.

(c) The couple is replaced by another pair of forces whose rotational effects are same.

Varignon's Theorem :

→ Next Page

Varignon's Theorem:



"In a concurrent force system, the moment of the resultant about a point is equal to the sum of the moments of the concurrent forces about the same point."

Proof :-

Consider the concurrent system of forces F_1 & F_2 originating from 'O'. Let 'R' be the resultant of the above two forces.

Consider the moment center 'P' along the horizontal and let the d_1, d_2 & d be the perpendicular distances of the forces F_1, F_2 & R respectively from the moment center 'P'.

Let θ_1, θ_2 & θ be the angles made by the forces F_1, F_2 & Resultant (R) with respect to the horizontal respectively.

As per the statement, we have to prove that

$$\boxed{F_1 d_1 + F_2 d_2 = Rd}$$

From Δ^{le} OCA

$$\sin\theta = \frac{CA}{OC}$$

$$\sin\theta = \frac{CB + BA}{R}$$

$$CB + BA = R\sin\theta \longrightarrow (a)$$

From Δ^{le} CQB

$$\sin\theta_1 = \frac{CB}{CQ} \quad \{\because CQ = F_1\}$$

$$\boxed{CB = F_1\sin\theta_1}$$

From fig, $BA = QD$

From Δ^{le} OQD

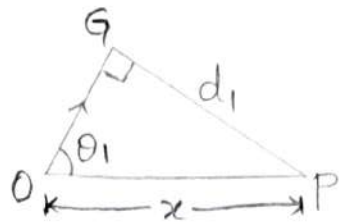
$$\sin\theta_2 = \frac{QD}{OQ} \quad \{\because OQ = F_2\}$$

$$\boxed{QD = BA = F_2\sin\theta_2}$$

Substituting the values of CB & BA in (a)

$$F_1\sin\theta_1 + F_2\sin\theta_2 = R\sin\theta. \longrightarrow (b)$$

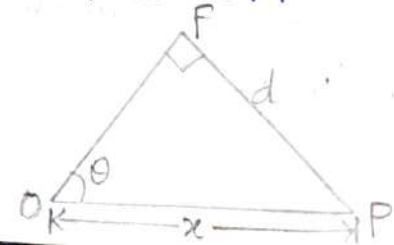
From Δ^{le} OGP



$$\sin\theta_1 = \frac{GP}{OP}$$

$$\boxed{\sin\theta_1 = \frac{d_1}{x}}$$

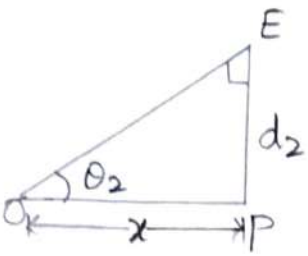
From Δ^{le} OFP



$$\sin\theta = \frac{FP}{OP}$$

$$\boxed{\sin\theta = \frac{d}{x}}$$

From Δ^{le} OEP



$$\sin\theta_2 = \frac{EP}{OP}$$

$$\boxed{\sin\theta_2 = \frac{d_2}{x}}$$

Substitute value of $\sin\theta_1$, $\sin\theta_2$ & $\sin\theta$ in (b)

$$F_1\left(\frac{d_1}{x}\right) + F_2\left(\frac{d_2}{x}\right) = R\left(\frac{d}{x}\right)$$

$$\frac{1}{x}\left[F_1d_1 + F_2d_2\right] = \frac{1}{x}\left[Rd\right]$$

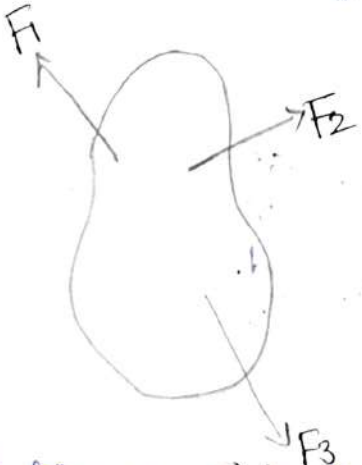
$$\boxed{F_1d_1 + F_2d_2 = Rd}$$

Hence proved

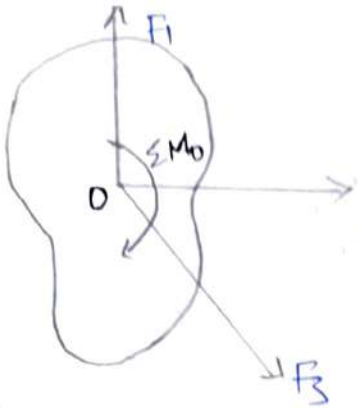
Resultant of Non-concurrent force system

Resultant of the force system is the one which will have the same rotational and translatory effect as the given system of forces is having.

> let F_1, F_2 & F_3 are the forces constituting on the body as shown in the figure.



> Each force can be replaced by a force of same magnitude acting in the same direction at Point 'O' and a moment about 'O'.

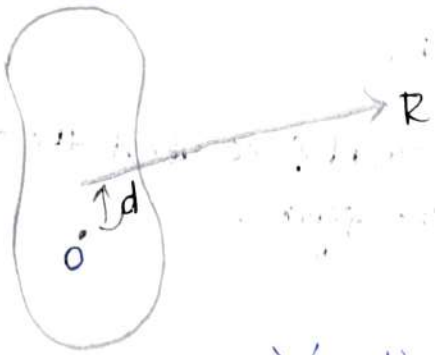


Where ΣM_O is the algebraic sum of the moments of the given forces about 'O'.

> At 'O' the concurrent forces F_1, F_2 & F_3 can be combined to get the resultant force (R). Now the resultant of the system is equal to the force 'R' at 'O' and a moment 'ΣM_O' as shown in the figure.



> The force 'R' and 'ΣM_O' can be replaced by a single force 'R' acting at a distance 'd' from 'O' such that the moment produced by this force about 'O' is equal to 'ΣM_O'



> Thus, we get a single force 'R' acting at a distance 'd' from the point 'O' which gives the same effect as the constitute forces of the system. Mathematically,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\alpha = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$$

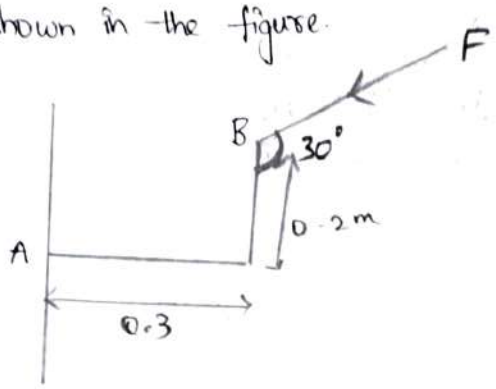
$$d = \frac{\Sigma M_O}{R}$$

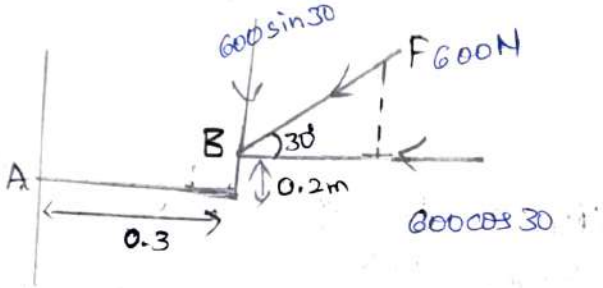
'x' intercept of 'R' $(x) = \frac{\Sigma M_O}{\Sigma V}$

'y' intercept of 'R' $(Y) = \frac{\Sigma M_O}{\Sigma H}$

Problems

① Find the moment of a moment of a force F = 600N about 'A' as shown in the figure.

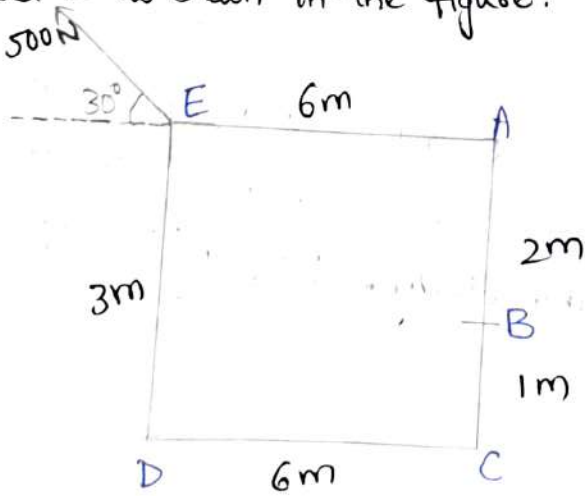




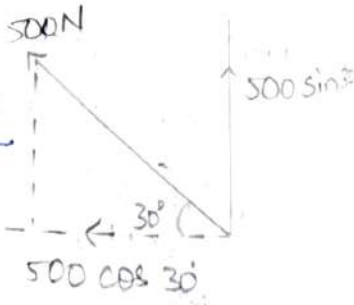
$M_A = (+600 \sin 30 \times 0.3) - 600 \cos 30 \times 0.2$

$M_A = -13.92 \text{ N-m} \ (\circlearrowleft)$

② Find the moment of 500N force about the points A, B, C and D as shown in the figure.



Sol



Moment about A, $M_A = 500 \sin 30 \times 6$

$= 1500 \text{ N-m} (\circlearrowright)$

Moment about B, $M_B = -500 \cos 30 \times 2 + 500 \sin 30 \times 6$

$= 633.97 \text{ N-m} (\circlearrowleft)$

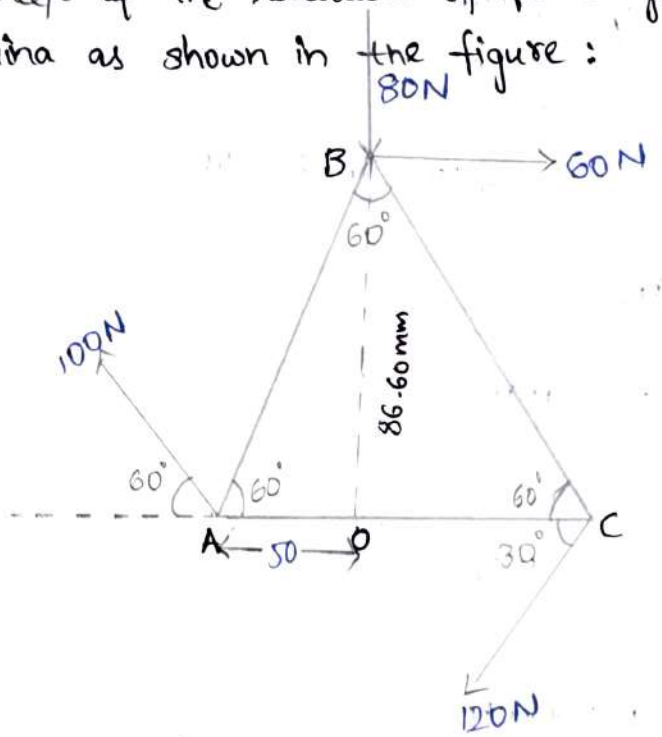
Moment about C, $M_C = -500 \cos 30 \times 3 + 500 \sin 30 \times 6$

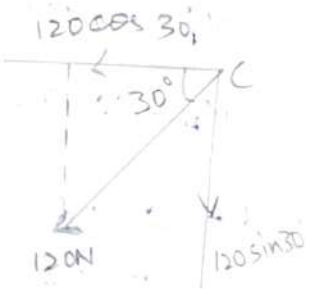
$= 200.96 \text{ N-m}$

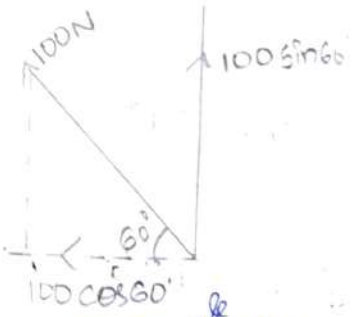
Moment about D, $M_D = -500 \cos 30 \times 3$

$= -1299.03 \text{ N-m} (\circlearrowright)$

③ Determine the magnitude of the resultant & x and y intercept of the resultant of force system acting on the lamina as shown in the figure:







Sol

$\Sigma V = -80 - 120 \sin 30 + 100 \sin 60$

$\boxed{\Sigma V = -53.39 \text{ N}}$

$\Sigma H = 60 - 120 \cos 30 - 100 \cos 60$

$\boxed{\Sigma H = -93.92 \text{ N}}$

From $\triangle^{le}$ ABO

$\tan 60 = \frac{BO}{AO}$

$\tan 60 = \frac{BO}{50}$

$\boxed{BO = 86.60 \text{ mm}}$

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$R = \sqrt{(-93.92)^2 + (-53.39)^2}$$

$$\boxed{R = 108.03\ N}$$

$$\alpha = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$$

$$\alpha = \tan^{-1}\left[\frac{-53.39}{-93.92}\right]$$

$$\boxed{\alpha = 29.59^\circ}$$

$$d = \frac{\Sigma M_A}{R}$$

$$x = \frac{\Sigma M_A}{\Sigma V} \qquad y = \frac{\Sigma M_A}{\Sigma H}$$

$$\Sigma M_A = 80 \times 50 + 60 \times 86.60 + 120 \sin 30 \times 100$$

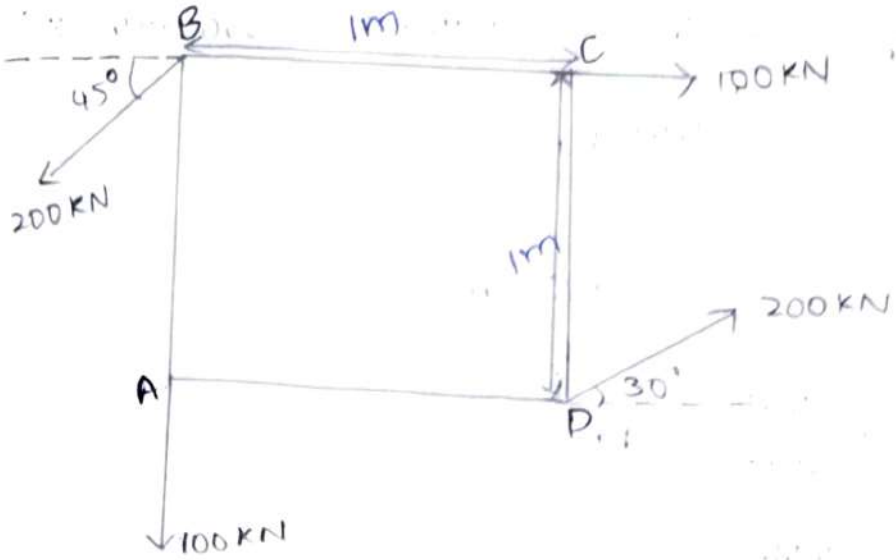
$$\boxed{\Sigma M_A = 15196\ N-m}$$

$$d = \frac{15196}{108.03} = 140.66\ mm$$

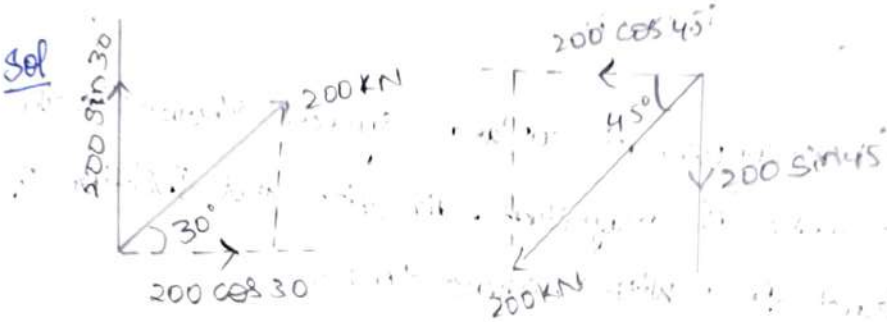
$$x = \frac{15196}{-53.39} = 284.62\ mm$$

$$y = \frac{15196}{-93.92} = 161.78\ mm$$

④ A rigid plate ABCD is subjected to forces as shown in the figure. Compute the magnitude, direction and line of action of the resultant of a system of forces with reference to the point 'A'.



Sol



$\Sigma V = -100 - 200\sin 45^\circ + 200\sin 30^\circ$

$\Sigma V = -141.42$ KN

$\Sigma H = -200\cos 45^\circ + 100 + 200\cos 30^\circ$

$\Sigma H = 131.78$ KN

$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$

$R = \sqrt{(131.78)^2 + (-141.42)^2}$

$R = 193.30$ KN

$\alpha = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$

$\alpha = \tan^{-1}\left[\frac{-141.42}{131.78}\right]$

$\alpha = 47.02^\circ$

$d = \frac{\Sigma M_A}{R}$ | $x = \frac{\Sigma M_A}{\Sigma V}$ | $y = \frac{\Sigma M_A}{\Sigma H}$

$$\Sigma M_n = -200\sin 30 \times 1 + 100 \times 1 - 200\cos 45 \times 1$$

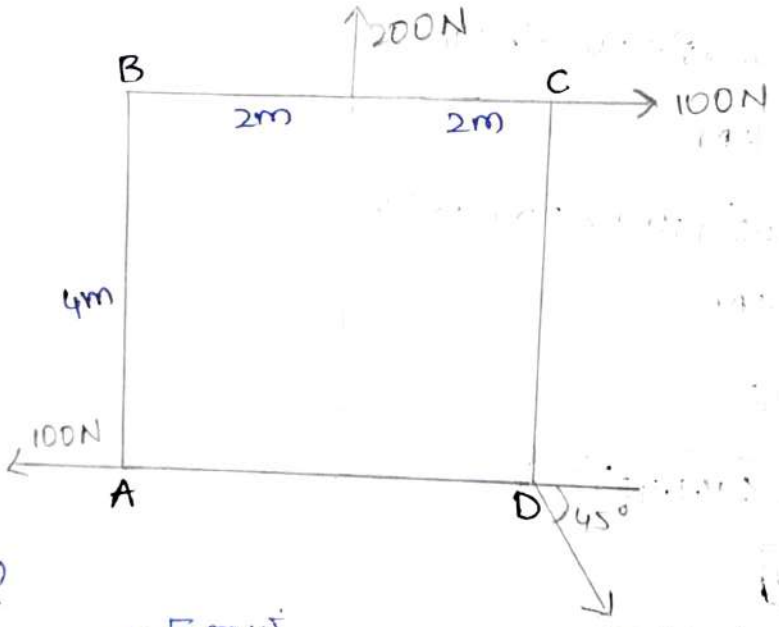
$$\boxed{\Sigma M_A = -141.42\ KN-m}$$

$$d = \frac{-141.42}{193.30} = 0.732\ m$$

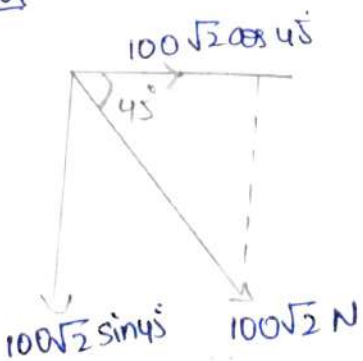
$$x = \frac{-141.42}{-141.42} = 1\ m$$

$$y = \frac{-141.42}{131.78} = 1.07\ m$$

⑤ For the non-concurrent coplanar system shown in the figure, determine the magnitude, direction and Position of the resultant force with reference to A.



Sol



$100\sqrt{2}$ N

$\Sigma H = 100 - 100 + 100\sqrt{2}\cos 45^\circ$

$\boxed{\Sigma H = 100\,N}$

$\Sigma V = 200 - 100\sqrt{2}\sin 45^\circ$

$\boxed{\Sigma V = 100\,N}$

$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$

$R = \sqrt{(100)^2 + (100)^2}$

$\boxed{R = 141.42\,N}$

$\alpha = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$

$\alpha = \tan^{-1}\left[\frac{100}{100}\right]$

$\alpha = \tan^{-1}(1)$

$\boxed{\alpha = 45^\circ}$

$\Sigma M_A = -200 \times 2 + 100 \times 4 + 100\sqrt{2}\sin 45^\circ \times 4$

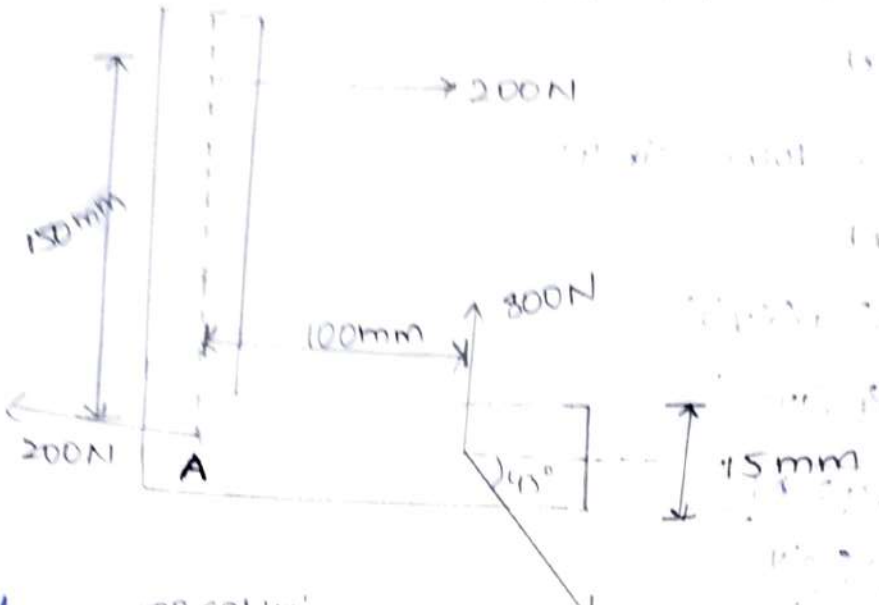
$\boxed{\Sigma M_A = 400\ N\text{-}m}$

$d = \frac{\Sigma M_A}{R} = \frac{400}{141.42} = 2.828m$

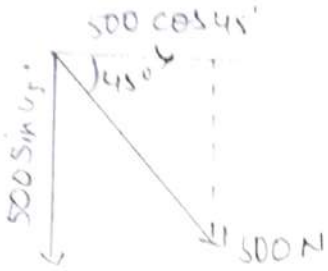
$X = \frac{\Sigma M_A}{\Sigma V} = \frac{400}{100} = 4m$

$Y = \frac{\Sigma M_A}{\Sigma H} = \frac{400}{100} = 4m$

⑥ Find the equilibrium force with respect to A as an origin for the system of forces shown in the figure.



Sol



$\Sigma H = 200 - 200 + 500\cos 45°$

$\Sigma H = 353.55\ N$

$\Sigma V = 800 - 500\sin 45°$

$\Sigma V = 446.44\ N$

$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$

$R = \sqrt{(353.55)^2 + (446.44)^2}$

$R = 569.47\ N$

$\alpha = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$

$\alpha = \tan^{-1}\left[\frac{446.44}{353.55}\right]$

$\alpha = 51.62°$

$\Sigma M_A = 200 \times 150 + 500 \sin 45^\circ \times 100 - 800 \times 100$

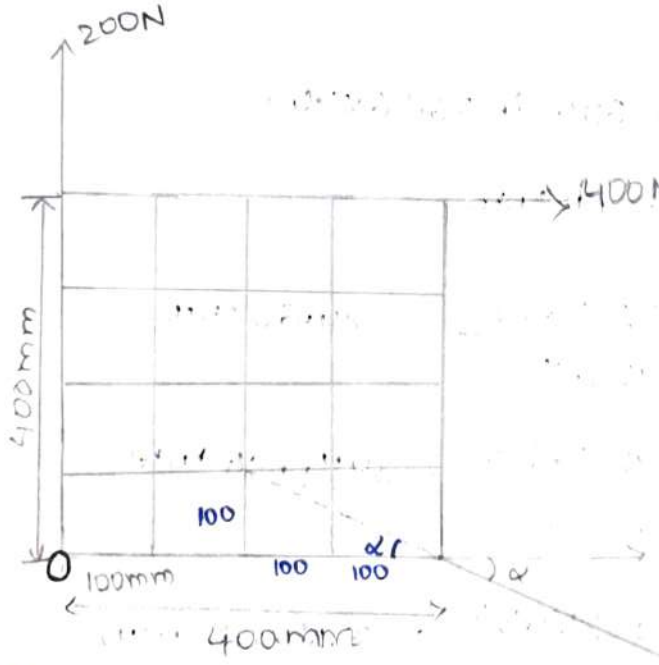
$\Sigma M_A = +14644.66 \text{ N-mm} \; (\curvearrowleft)$ Anti-clockwise.

$$d = \frac{\Sigma M_A}{R} = \frac{14644.66}{569.497} = 25.71 \text{ mm}$$

$$x = \frac{\Sigma M_A}{\Sigma V} = \frac{14644.66}{446.44} = 32.80 \text{ mm}$$

$$y = \frac{\Sigma M_A}{\Sigma H} = \frac{14644.66}{353.55} = 41.42 \text{ mm}$$

(7) Determine the magnitude, direction of the resultant for the force system acting on the lamina as shown in the figure. Also find the position of the resultant.

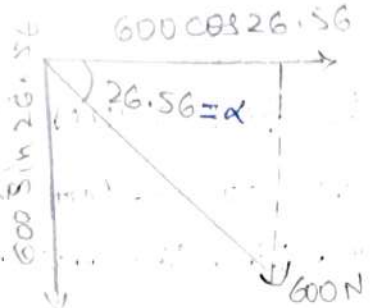


Sol

$$\tan\alpha = \frac{100}{200}$$

$$\alpha = \tan^{-1}\left[\frac{100}{200}\right]$$

$\alpha = 26.56^\circ$



$\Sigma H = 600\cos 26.56 + 400$

$\Sigma H = 936.67\ N$

$\Sigma V = 200 - 600\sin 26.56$

$\Sigma V = -68.28\ N$

$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$

$R = \sqrt{(936.67)^2 + (-68.28)^2}$

$R = 939.15\ N$

$\alpha = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$

$\alpha = \tan^{-1}\left[\frac{-68.28}{936.67}\right]$

$\alpha = -4.17^\circ$

$\Sigma M_O = 400 \times 400 + 600\sin 26.56 \times 400$

$\Sigma M_O = 267312.33\ N\text{-}mm$

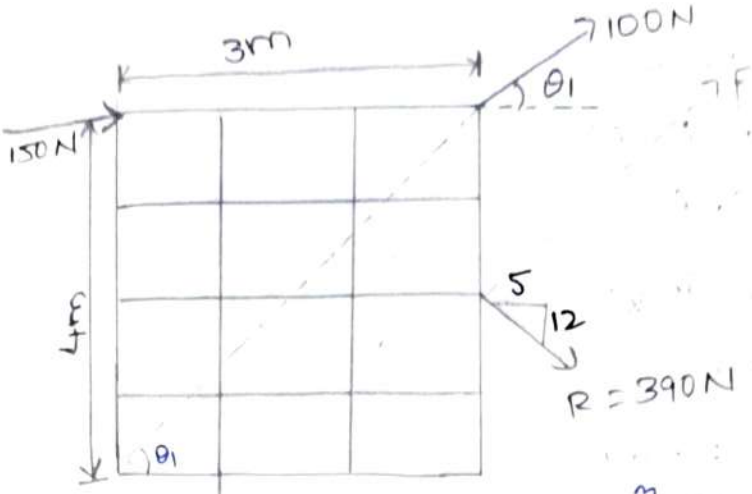
$d = \frac{\Sigma M_O}{R} = \frac{267312.33}{939.15} = 284.63\ mm$

$x = \frac{\Sigma M_O}{\Sigma V} = \frac{267312.33}{\cancel{-}68.28} = 3914.94\ mm$

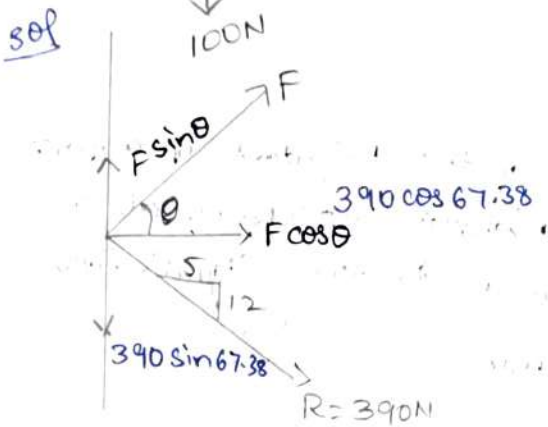
Neglected

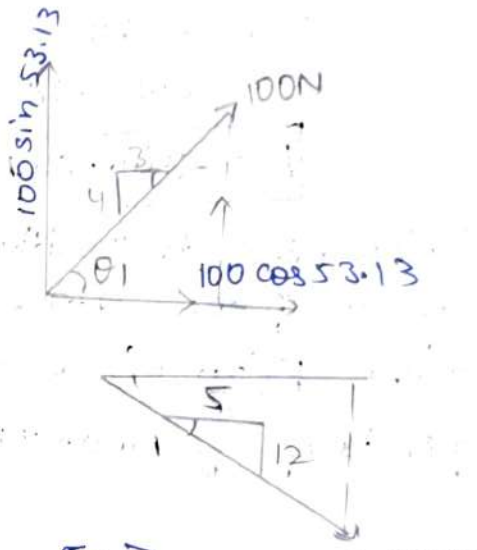
$y = \frac{\Sigma M_O}{\Sigma H} = \frac{267312.33}{936.67} = 285.38\ mm$

⑧ The force 390N shown in figure is the resultant of four forces. Out of them, three are shown in the figure. Find the missing force and its position.



Sol





$\tan\theta_1 = \frac{4}{3}$

$\theta_1 = \tan^{-1}\left[\frac{4}{3}\right]$

$\theta_1 = 53.13°$

$\tan\theta_2 = \left[\frac{12}{5}\right]$

$\theta_2 = \tan^{-1}\left[\frac{12}{5}\right]$

$\theta_2 = 67.38°$

$\Sigma H = R$

$100\cos 53.13 + F\cos\theta + 150 = 390\cos 67.38$

$F\cos\theta = -60 \longrightarrow ①$

$\Sigma V = R$

$100\sin 53.13 - 100 + F\sin\theta = -390\sin 67.38$

$F\sin\theta = -340 \longrightarrow ②$

Dividing Eq^n ② by Eq^n ①

$$\frac{F\sin\theta}{F\cos\theta} = \frac{-340}{-60}$$

$$\tan\theta = 5.66$$

$$\theta = \tan^{-1}[5.66]$$

$$\boxed{\theta = 79.98^\circ}$$

Substitute 'θ' in (1)

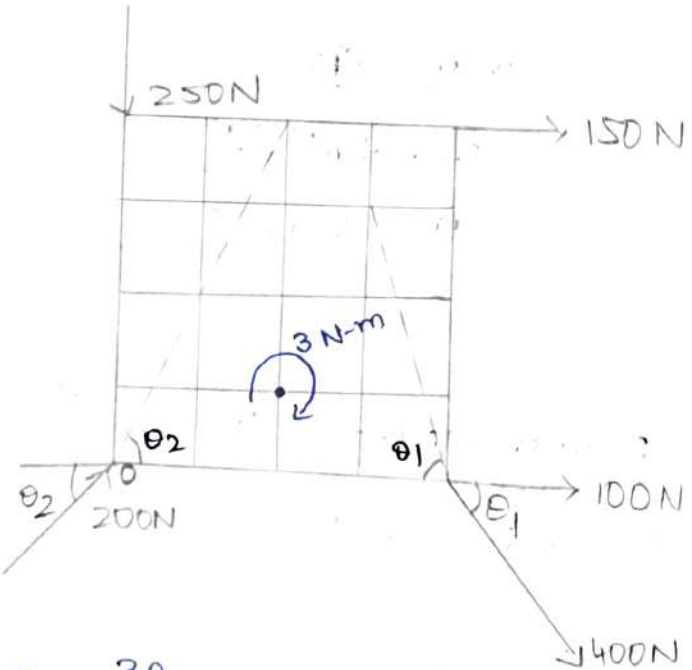
$$F\cos\theta = -60$$

$$F\cos(79.98) = -60$$

$$F = \frac{-60}{\cos 79.98}$$

$$\boxed{F = -344.84\ N}$$

(9) Determine the position of the resultant of a force system if the external moment $M = 3\ N\text{-}m$ at a point $P(20, 10)$ as shown in the figure. Each grid is of size 10mm x 10mm.



Sol

$$\tan\theta_1 = \frac{30}{10}$$

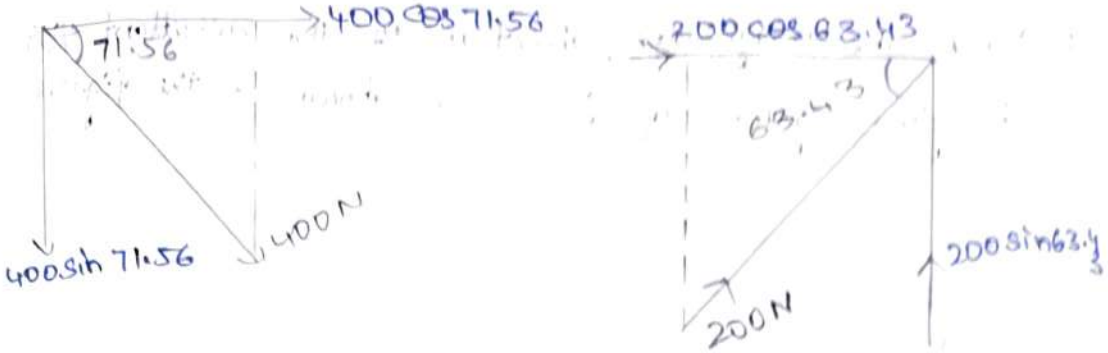
$$\theta_1 = \tan^{-1}\left[\frac{30}{10}\right]$$

$$\theta_1 = 71.56^\circ$$

$$\tan\theta_2 = \frac{40}{20}$$

$$\theta_2 = \tan^{-1}\left[\frac{40}{20}\right]$$

$$\theta_2 = 63.43^\circ$$



$$\Sigma H = 150 + 100 + 400 \cos 71.56 + 200 \cos 63.43$$

$$\Sigma H = 465.98 \text{ N}$$

$$\Sigma V = -250 - 400 \sin 71.56 + 200 \sin 63.43$$

$$\Sigma V = -450.58 \text{ N}$$

$$R = \sqrt{(\Sigma V)^2 + (\Sigma H)^2}$$

$$R = \sqrt{(465.98)^2 + (-450.58)^2}$$

$$\boxed{R = 648.197 \text{ N}}$$

$$\alpha = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$$

$$\alpha = \tan^{-1}\left[\frac{-450.58}{465.98}\right]$$

$$\boxed{\alpha = -44.03^\circ}$$

$$\Sigma M_O = 400 \sin 71.56 \times 40 + 150 \times 40 + 3 \times 20$$

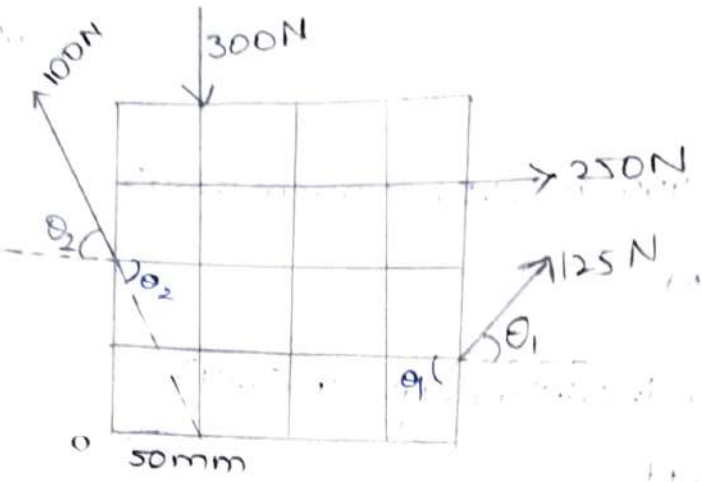
$$\boxed{\Sigma M_O = 21238.48 \text{ N-mm}}$$

$$d = \frac{\Sigma M_O}{R} = \frac{21238.48}{648.197} = 32.76 \text{ mm}$$

$$X = \frac{\Sigma M_O}{\Sigma V} = \frac{21238.48}{-450.58} = -47.13 \text{ mm}$$

$$Y = \frac{\Sigma M_O}{\Sigma H} = \frac{21238.48}{465.98} = 45.57 \text{ mm}$$

(10) Find the magnitude, direction of the resultant for the given system of forces as shown in the figure.



Sol

$\tan\theta_1 = \frac{50}{50}$

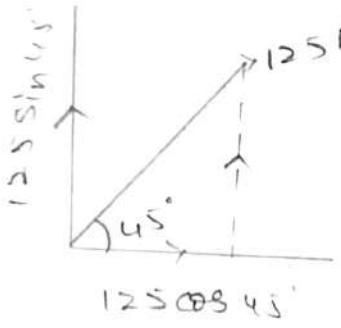
$\theta_1 = \tan^{-1}\left[\frac{50}{50}\right]$

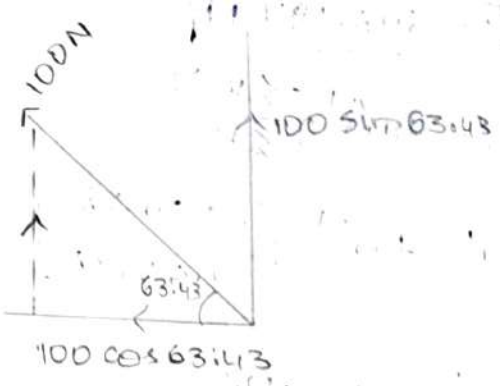
$\theta_1 = 45^\circ$

$\tan\theta_2 = \frac{100}{50}$

$\theta_2 = \tan^{-1}\left[\frac{100}{50}\right]$

$\theta_2 = 63.43^\circ$





$\Sigma H = 125\cos 45 - 100\cos 63.43 + 250$

$\Sigma H = 293.65\ N$

$\Sigma V = 125\sin 45 + 100\sin 63.43 - 300$

$\Sigma V = -122.17\ N$

$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$

$R = \sqrt{(293.65)^2 + (-122.17)^2}$

$\boxed{R = 318.05\ N}$

$$\alpha = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$$

$$\alpha = \tan^{-1}\left[\frac{-122.17}{293.65}\right]$$

$$\boxed{\alpha = -22.58^\circ}$$

$$\Sigma M_O = 300 \times 50 + 250 \times 150 + 125\cos 45 \times 50 - 125\sin 45 \times 200 - 100\cos 63.43 \times 100$$

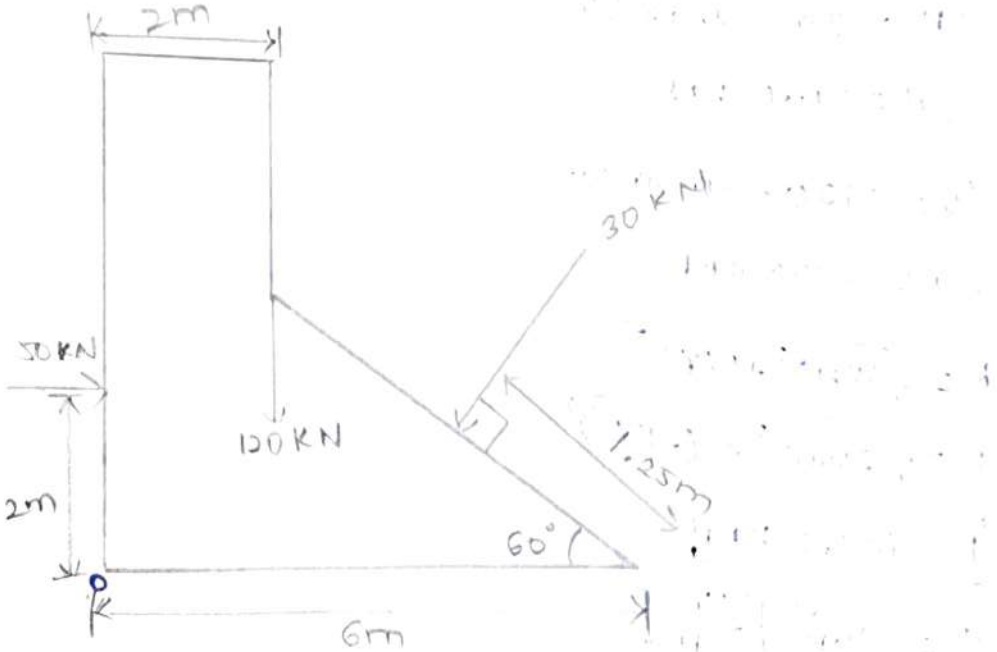
$$\Sigma M_O = 34768.83 \text{ N-m}$$

$$d = \frac{\Sigma M_O}{R} = \frac{34768.83}{318.05} = 109.31 \text{ mm}$$

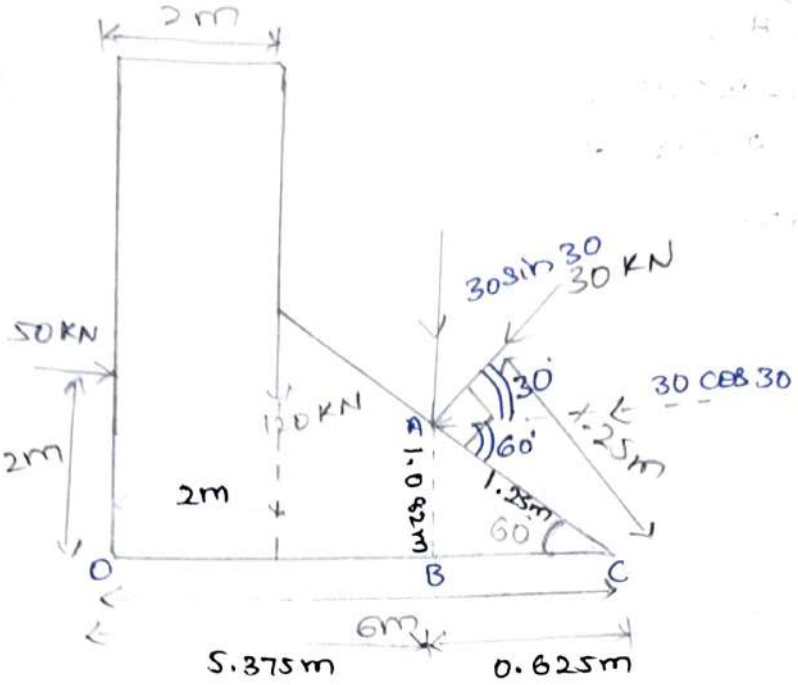
$$x = \frac{\Sigma M_O}{\Sigma V} = \frac{34768.83}{-122.17} = -284.59 \text{ mm}$$

$$y = \frac{\Sigma M_O}{\Sigma H} = \frac{34768.83}{293.65} = 118.40 \text{ mm}$$

(11) forces acting on 1 meter length of a dam are as shown in the figure. Determine the resultant force acting on the dam. Calculate the point of intersection of resultant with the base.



Sol



From Δ^{le} ABC

$$\sin 60^\circ = \frac{AB}{AC} \qquad \cos 60^\circ = \frac{BC}{AC}$$

$$AB = AC \sin 60^\circ \qquad BC = 1.25 \times \cos 60^\circ$$

$$AB = 1.25 \times \sin 60^\circ \qquad \boxed{BC = 0.625\ m}$$

$$\boxed{AB = 1.082\ m}$$

$$\Sigma H = 50 - 30 \cos 30$$

$$\Sigma H = 24.01\ KN$$

$$\Sigma V = -120 - 30 \sin 30$$

$$\Sigma V = -135\ KN$$

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$R = \sqrt{(24.01)^2 + (-135)^2}$$

$$\boxed{R = 137.1\ KN}$$

$$\alpha = \tan^{-1}\left[\frac{\Sigma V}{\Sigma H}\right]$$

$$\alpha = \tan^{-1}\left[\frac{-135}{24.01}\right]$$

$$\boxed{\alpha = -79.9^\circ}$$

$$\Sigma M_O = 50 \times 2 + 30 \sin 30 \times 5.375 - 30 \cos 30 \times 1.082 + 120 \times 7$$

$$\boxed{\Sigma M_O = 392.51 \text{ KN-m}}$$

$$\bar{x} = \frac{\Sigma M_O}{\Sigma V} = \frac{392.51}{-135} = -2.90 \text{ m}$$

$$\bar{y} = \frac{\Sigma M_O}{\Sigma H} = \frac{392.51}{24.01} = 16.34 \text{ m}$$

$$d = \frac{\Sigma M_O}{R} = \frac{392.51}{137.1} = 2.86 \text{ m}$$